

# lab04

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```
# Q1
```

```
library(readxl)
library(tidyverse)
```

```
Warning: pacote 'tidyverse' foi compilado no R versão 4.5.3
```

```
Warning: pacote 'ggplot2' foi compilado no R versão 4.5.3
```

```
Warning: pacote 'dplyr' foi compilado no R versão 4.5.3
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.1      ✓ readr      2.1.5
✓ forcats    1.0.1      ✓ stringr    1.5.2
✓ ggplot2    4.0.2      ✓ tibble     3.3.0
✓ lubridate  1.9.4      ✓ tidyr      1.3.1
✓ purrr      1.1.0
— Conflicts — tidyverse_conflicts()
—
* dplyr::filter() masks stats::filter()
* dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
# Q2
```

```
fname <- "WDIEXCEL.xlsx"

file.exists(fname)
```

```
[1] TRUE
```

```
# Q3

countries <- read_excel(
  fname,
  sheet = "Country"
)

countries
```

```
# A tibble: 264 × 31
  `Country Code` `Short Name`      `Table Name` `Long Name` `2-alpha code`
  <chr>          <chr>          <chr>        <chr>      <chr>
1 ABW          Aruba          Aruba        Aruba      AW
2 AFE          Africa Eastern and So... Africa East... Africa Eas... ZH
3 AFG          Afghanistan      Afghanistan  Islamic St... AF
4 AFW          Africa Western and Ce... Africa West... Africa Wes... ZI
5 AGO          Angola          Angola       People's R... AO
6 ALB          Albania          Albania      Republic o... AL
7 AND          Andorra          Andorra      Principali... AD
8 ARB          Arab World      Arab World   Arab World   1A
9 ARE          United Arab Emirates  United Arab... United Ara... AE
10 ARG         Argentina      Argentina    Argentine ... AR
# i 254 more rows
# i 26 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, ...
```

```
glimpse(countries)
```

```
Rows: 264
Columns: 31
$ `Country Code`      <chr> "ABW", "AFE", "AFG...
$ `Short Name`        <chr> "Aruba", "Africa E...
$ `Table Name`        <chr> "Aruba", "Africa E...
$ `Long Name`         <chr> "Aruba", "Africa E...
$ `2-alpha code`      <chr> "AW", "ZH", "AF", ...
$ `Currency Unit`     <chr> "Aruban florin", N...
$ `Special Notes`     <chr> NA, "26 countries,...
```

```

$ Region <chr> "Latin America & C...
$ `Income Group` <chr> "High income", NA,...
$ `WB-2 code` <chr> "AW", "ZH", "AF", ...
$ `National accounts base year` <chr> "2013", NA, "2016"...
$ `National accounts reference year` <dbl> NA, NA, NA, NA, NA...
$ `SNA price valuation` <chr> "Value added at ba...
$ `Lending category` <chr> NA, NA, "IDA", NA,...
$ `Other groups` <chr> NA, NA, "HIPC", NA...
$ `System of National Accounts` <chr> "Country uses the ...
$ `Alternative conversion factor` <lgl> NA, NA, NA, NA, NA...
$ `PPP survey year` <lgl> NA, NA, NA, NA, NA...
$ `Balance of Payments Manual in use` <chr> "BPM6", NA, "BPM6"...
$ `External debt Reporting status` <chr> NA, NA, "Estimate"...
$ `System of trade` <chr> "General trade sys...
$ `Government Accounting concept` <chr> NA, NA, "Consolida...
$ `IMF data dissemination standard` <chr> "Enhanced General ...
$ `Latest population census` <chr> "2020 (expected)",...
$ `Latest household survey` <chr> NA, NA, "Multiple ...
$ `Source of most recent Income and expenditure data` <chr> NA, NA, "Integrate...
$ `Vital registration complete` <chr> "Yes", NA, NA, NA,...
$ `Latest agricultural census` <dbl> NA, NA, 2003, NA, ...
$ `Latest industrial data` <dbl> NA, NA, NA, NA, NA...
$ `Latest trade data` <dbl> 2018, NA, 2018, NA...
$ `Latest water withdrawal data` <lgl> NA, NA, NA, NA, NA...

```

```

countries %>%
  select(`Short Name`, `Country Code`)

```

```

# A tibble: 264 × 2
  `Short Name`      `Country Code`
  <chr>            <chr>
1 Aruba            ABW
2 Africa Eastern and Southern AFE
3 Afghanistan      AFG
4 Africa Western and Central AFW
5 Angola            AGO
6 Albania           ALB
7 Andorra           AND
8 Arab World        ARB
9 United Arab Emirates ARE
10 Argentina        ARG
# i 254 more rows

```

```
# Q4

countries <- countries %>%
  filter(
    `Short Name` %in% c(
      "Brazil",
      "Argentina",
      "Chile",
      "Mexico",
      "United States"
    )
  )
```

```
# Q5

WDI <- read_excel(
  fname,
  sheet = "Data"
)
```

```
glimpse(WDI)
```

```
Rows: 396,970
Columns: 70
$ `Country Name`    <chr> "Africa Eastern and Southern", "Africa Eastern and So...
$ `Country Code`    <chr> "AFE", "AFE", "AFE", "AFE", "AFE", "AFE", "AFE", "AFE...
$ `Indicator Name`  <chr> "Access to clean fuels and technologies for cooking (...
$ `Indicator Code`  <chr> "EG.CFT.ACCS.ZS", "EG.CFT.ACCS.RU.ZS", "EG.CFT.ACCS.U...
$ `1960`            <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1961`            <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1962`            <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1963`            <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1964`            <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1965`            <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1966`            <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1967`            <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1968`            <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1969`            <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1970`            <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1971`            <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ `1972`            <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
```

|           |                                                                    |
|-----------|--------------------------------------------------------------------|
| \$ `1973` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1974` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1975` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1976` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1977` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1978` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1979` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1980` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1981` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1982` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1983` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1984` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1985` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1986` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1987` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1988` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1989` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1990` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1991` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1992` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1993` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1994` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1995` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1996` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1997` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1998` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `1999` | <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N... |
| \$ `2000` | <dbl> 1.149470e+01, 3.548167e+00, 3.235069e+01, 1.991726e+0...     |
| \$ `2001` | <dbl> 1.184106e+01, 3.704024e+00, 3.268334e+01, 1.994428e+0...     |
| \$ `2002` | <dbl> 1.220252e+01, 3.890043e+00, 3.310018e+01, 2.154865e+0...     |
| \$ `2003` | <dbl> 1.257338e+01, 4.091720e+00, 3.349046e+01, 2.249322e+0...     |
| \$ `2004` | <dbl> 1.295869e+01, 4.274249e+00, 3.399118e+01, 2.372445e+0...     |
| \$ `2005` | <dbl> 1.337114e+01, 4.532521e+00, 3.447061e+01, 2.344954e+0...     |
| \$ `2006` | <dbl> 1.378271e+01, 4.757490e+00, 3.492266e+01, 2.517248e+0...     |
| \$ `2007` | <dbl> 1.425418e+01, 5.004035e+00, 3.550103e+01, 2.679514e+0...     |
| \$ `2008` | <dbl> 1.473659e+01, 5.248924e+00, 3.591535e+01, 2.590885e+0...     |
| \$ `2009` | <dbl> 1.517156e+01, 5.499747e+00, 3.643522e+01, 2.618785e+0...     |
| \$ `2010` | <dbl> 1.562017e+01, 5.747851e+00, 3.682269e+01, 2.741495e+0...     |
| \$ `2011` | <dbl> 1.612320e+01, 6.028196e+00, 3.725257e+01, 2.890976e+0...     |
| \$ `2012` | <dbl> 1.659973e+01, 6.309218e+00, 3.763599e+01, 3.167343e+0...     |
| \$ `2013` | <dbl> 1.706105e+01, 6.643625e+00, 3.796932e+01, 3.172357e+0...     |
| \$ `2014` | <dbl> 1.763415e+01, 6.932310e+00, 3.834023e+01, 3.187196e+0...     |
| \$ `2015` | <dbl> 1.814583e+01, 7.259936e+00, 3.874199e+01, 3.392228e+0...     |

```

$ `2016`      <dbl> 1.868512e+01, 7.606712e+00, 3.905263e+01, 3.885960e+0...
$ `2017`      <dbl> 1.920563e+01, 7.926604e+00, 3.932107e+01, 4.022374e+0...
$ `2018`      <dbl> 1.974277e+01, 8.309896e+00, 3.964953e+01, 4.303507e+0...
$ `2019`      <dbl> 2.033268e+01, 8.704591e+00, 3.996830e+01, 4.439086e+0...
$ `2020`      <dbl> 2.086280e+01, 9.106640e+00, 4.035463e+01, 4.628237e+0...
$ `2021`      <dbl> 2.141962e+01, 9.480804e+00, 4.072381e+01, 4.812721e+0...
$ `2022`      <dbl> 21.996456, 9.903209, 41.026351, 48.742043, 33.998179,...
$ `2023`      <dbl> 22.54144, 10.28815, 41.28997, 50.66752, 35.24024, 80...
$ `2024`      <dbl> NA, NA, NA, 52.59817, 36.08057, 80.04927, NA, NA, NA,...
$ `2025`      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...

```

```
# Q6
```

```

WDI <- WDI %>%
  select(
    `Country Code`,
    `Indicator Code`,
    `1960`:`2018`
  )

```

```
# Q7
```

```

WDI <- WDI %>%
  pivot_longer(                                #coloca em formato tidy
    cols = `1960`:`2018`,
    names_to = "YEAR",
    values_to = "VALUE"
  )

```

```
head(WDI)
```

```

# A tibble: 6 × 4
  `Country Code` `Indicator Code` YEAR  VALUE
  <chr>          <chr>          <chr> <dbl>
1 AFE           EG.CFT.ACCS.ZS  1960    NA
2 AFE           EG.CFT.ACCS.ZS  1961    NA
3 AFE           EG.CFT.ACCS.ZS  1962    NA
4 AFE           EG.CFT.ACCS.ZS  1963    NA
5 AFE           EG.CFT.ACCS.ZS  1964    NA
6 AFE           EG.CFT.ACCS.ZS  1965    NA

```

```
# Q8
```

```
WDI <- WDI %>%  
  mutate(  
    YEAR = as.integer(YEAR)  
  )  
  
class(WDI$YEAR)
```

```
[1] "integer"
```

```
str(WDI$YEAR)
```

```
int [1:23421230] 1960 1961 1962 1963 1964 1965 1966 1967 1968 1969 ...
```

```
# Q9
```

```
WDI <- WDI %>%  
  semi_join(  
    countries,  
    by = "Country Code"  
  ) %>%  
  left_join(  
    countries %>%  
      select(  
        `Country Code`,  
        `Short Name`  
      ),  
    by = "Country Code"  
  ) %>%  
  filter(  
    `Indicator Code` == "SP.DYN.CBRT.IN"  
  ) %>%  
  select(  
    `Country Code`,  
    `Short Name`,  
    YEAR,  
    `VALUE`  
  ) %>%  
  rename(  
    Country = `Country Code`,  
    Name = `Short Name`,  
    Value = `VALUE`  
  )
```

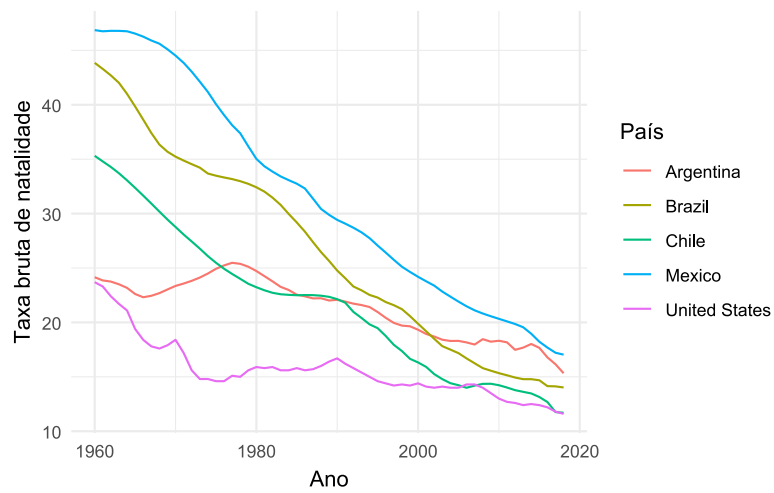
```
`SP.DYN.CBRT.IN` = VALUE  
)
```

```
# Q10
```

```
WDI <- WDI %>%  
  arrange(  
    `Country Code`,  
    YEAR  
  )
```

```
# Q11
```

```
ggplot(  
  WDI,  
  aes(  
    x = YEAR,  
    y = `SP.DYN.CBRT.IN`,  
    color = `Short Name`,  
    group = `Short Name`  
  )  
) +  
  geom_line() +  
  labs(  
    x = "Ano",  
    y = "Taxa bruta de natalidade",  
    color = "País"  
  ) +  
  theme_minimal()
```



```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-08 22:49:59 -03"
```